Scientific Audit, Negative-Result Repair, and Next-Iteration Strategy for GraphKAN AMO NQS: Stage 4 N=8 Five-Seed Replication

Prepared for supervisory review

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Abstract

This report audits the uploaded Stage 4 package for the GraphKAN AMO neural-quantum-state project. The package reports a convention-locked NetKet/Flax hard-point transfer gate at $N = 8$, $\Omega = 1$, $\Delta = 2$, $C_6 = 4$ comparing five 21-parameter pair ansatz variants over five matched seeds. The strongest empirical result is narrow but real: the RBF endpoint-sum model improves exact all-state energy error, density mean absolute error (MAE), and connected-correlation MAE relative to the matched RBF radial model in all five paired seeds. The median exact changes are -0.364296, -0.057812, and -0.006793, respectively. This is a meaningful repair signal for the previously identified open-chain hard regime. It is not a general KAN-superiority result, not a Graph-KAN result, not a geometry-transfer result, and not manuscript-ready evidence. The next action is not blind scale-up; it is a Stage 4B capacity/compute control gate with stronger MLP/Jastrow baselines, cleaned gate definitions, and removal of stale contradictory artifacts. If those gates pass, a bolder hard-regime patch and finite-size campaign is scientifically justified within laptop/GPU/Colab-scale resources.

Audit scope	Student prompt, prior Stage 3E supervisory audit, Stage 4 code ZIP, Stage 4 result tarball, SHA-256 sidecar, configs, model code, tests, raw metrics, exact site/correlation observables, clean replay log, environment metadata, and aggregation scripts.
Resource assumptions	Personal/team CPU machines, optional laptop GPUs up to RTX 3080 Ti class, 6–16 GB VRAM, Colab Free where useful, no paid cloud/HPC/QPU requirement.
Primary decision use	Decide whether Stage 4 authorises N=10+, Graph-KAN, geometry transfer, manuscript drafting, or only targeted controls and repair.

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1 Executive Verdict and Strategic Position

Amber verdict: promising hard-point repair, but scale-up requires targeted diagnostics

Approve Stage 4 as an internal hard-regime transfer diagnostic. Do not approve $N \geq 10$ scale-up, GraphKAN, geometry transfer, or manuscript drafting yet. The submitted evidence supports a narrow statement: at the single open-chain hard point $N = 8$, $\Omega = 1$, $\Delta = 2$, $C_6 = 4$, the 21-parameter RBF endpoint-sum pair ansatz improves exact post-training energy, density, and connected-correlation errors relative to the matched 21-parameter RBF radial pair ansatz over five paired seeds. The evidence does not yet establish a method-level advantage over strong baselines, because the comparator is itself a failed radial control and the submitted Stage 4 result table contains no capacity-swept MLP, smooth/RBF Jastrow, RBM, DMRG, geometry-transfer, or larger-size controls.

Current evidence verdict. The current result is a replicated, local, architecture-diagnostic positive. It is stronger than Stage 3E because the preferred endpoint-sum variant improves all three tracked exact observables relative to RBF radial at $N = 8$, whereas the earlier full-endpoint result at $N = 6$ improved energy/density but carried a small correlation penalty. It is still one Hamiltonian instance and five seeds.

Next-iteration approval. Approve a focused Stage 4B: N=8 hard-point fairness controls and artifact/gate cleanup. Do not allow a broad campaign until the endpoint-sum result survives stronger controls.

Long-term publishability potential. The project remains strategically alive. The high-upside path is not a generic KAN paper. The plausible paper is a Rydberg hard-regime boundary-feature/geometry-aware pair-ansatz study with exact observables, failure maps, and fair controls. A top-journal route remains conditional: it would require persistent hard-regime repair across an adjacent parameter patch, finite-size transfer, strong baselines, and a physical explanation of why the endpoint-sum feature works.

Strongest evidence

- The Stage 4 result archive hash matches the supplied SHA-256 sidecar.
- The submitted clean replay log records a successful NQS environment test/preflight/convention replay; the local static audit passed all non-optional tests.
- All 25 scheduled Stage 4 VMC rows completed.
- The dense-versus-NetKet operator discrepancy is at machine precision in all raw rows: maximum matrix discrepancy 3.55×10^{-15} and maximum spectral discrepancy 6.39×10^{-14}.
- RBF endpoint-sum is 5/5 valid and improves exact energy, density, and connected-correlation errors relative to RBF radial in every paired seed.

Weakest evidence

- The main comparator, RBF radial, is 0/5 valid. Improvement over it is a repair signal, not a strong-baseline victory.
- No MLP or smooth/RBF Jastrow capacity/compute sweep appears in the submitted Stage 4 metrics.
- The result is one open-chain $N = 8$ Hamiltonian point selected because earlier grids identified it as hard.
- The raw files use Stage 3F-compatible names and include a stale `n8_joint_gate.csv` reporting `NOT_APPLICABLE`, while the current `n8_joint_gates.csv` reports the intended dual-pass gates.
- The runner's scientific-health classification uses sampled density/correlation errors, while the headline analysis uses exact all-state post-training observables. This does not overturn the endpoint-sum result, but it must be made explicit before scaling.

One-paragraph next action. The student should run **Stage 4B: N=8 Hard-Point Fairness and Gate-Definition Audit**. Before more physics runs, remove stale aggregate files, rename Stage 4 raw outputs or document their compatibility status, add explicit `sample_health_validity` and `exact_health_validity` columns, and add a Stage 4 schema test. Then run capacity-/compute-swept MLP-pair, endpoint-MLP, smooth/RBF Jastrow and endpoint-sum/RBF-radial controls at the same hard point. Only if endpoint-sum is not dominated should the project launch the adjacent hard-regime grid and finite-size transfer ladder.

2 Uploaded Material and Provenance

2.1 Material inspected

The audit inspected the following uploaded sources: the Stage 4 code ZIP, the Stage 4 result tarball, the Stage 4 SHA-256 sidecar, the prior Stage 3E supervisory audit PDF, and the uploaded instruction/student-note text. The extracted code package contains a production-style repository with accumulated Stage 2–4 scripts, configs, tests, documentation and model code. The extracted result package contains the Stage 4 N=8 metrics, histories, exact site observables, exact connected-correlation observables, aggregate tables, preflight metadata, run manifest, clean replay log and environment file.

Table 1: File-inventory summary from original uploaded archives. Cache/platform files include AppleDouble `._` files, `__MACOSX`, `__pycache__`, and compiled bytecode.

Package	Files	Bytes	Central files	Cache/platform files
code ZIP	310	7,278,621	12	4
results tarball	41	804,862	14	21
prior Stage 3E audit PDF	1	457,943	0	0
student prompt	1	44,110	0	0
SHA-256 sidecar	1	120	0	0

Central files inspected include `STAGE_4_README.md`, `configs/netket_stage4_n8_five_seed_replication.yaml`, `scripts/run_stage4.py`, `scripts/run_stage3f.py`, `scripts/summarize_stage4.py`, `scripts/run_stage4_clean_replay.`
`src/graphkan_amo_nqs/models/flax_nqs.py`, `tests/test_basis_family_dispatch.py`, `raw/stage3f_metrics.csv`, `raw/stage3f_site_observables.csv`, `raw/stage3f_corr_observables.csv`, `aggregated/model_summary.csv`, `aggregated/paired_contrasts.csv`, `aggregated/n8_joint_gates.csv`, `environment/nqs_preflight.json`, `environment/run_manifest.json`, and `clean_replay/stage4_clean_replay.log`.

2.2 Provenance reconstruction

The iteration sequence is coherent. Stage 2.6 identified the difficult $C_6 = 4$, high-detuning regime. Stage 3B suggested an endpoint-aware repair but had a basis-family ambiguity. Stage 3E corrected taxonomy and found a replicated $N = 6$ result: RBF endpoint improved energy and density relative to RBF radial, but not connected correlations. The Stage 3E audit therefore recommended an $N = 8$ hard-point transfer gate with stronger controls before larger scale-up. Stage 4 implements the first part of that request: it freezes a 21-parameter $N = 8$ hard-point comparison over five seeds and adds a new endpoint-sum candidate. It does not yet implement the stronger MLP/Jastrow capacity-control part.

Table 2: Reconstructed Stage 4 run inventory from `raw/stage3f_metrics.csv`.

Quantity	Value
Scheduled rows	25
Completed rows	25
Failed rows	0
Valid rows	14
Flagged rows	11
Models	5
Seeds	0,1,2,3,4
System size	$N = 8$
Hamiltonian point	$\Omega = 1$, $\Delta = 2$, $C_6 = 4$
Samples per run	4096
SR diagonal shift	0.1
Instance ID	`ryd_chain_N8_D2_C64_4d66847e71a41384`

2.3 Missing, stale, or suspicious artifacts

The package is adequate for internal review but not for artifact evaluation. The most important issues are: (i) no saved model checkpoints/parameters; (ii) no clean replay transcript that explicitly reruns the 25 VMC trajectories; (iii) Stage 4 result files are still named `stage3f_*`; (iv) a stale `n8_joint_gate.csv` file reports `NOT_APPLICABLE` while `n8_joint_gates.csv` gives the intended dual pass; and (v) the tarball includes many AppleDouble `._` files. These are repairable, but they matter because the project is now moving from diagnostics toward a publishability path.

3 Student Claim vs Evidence Audit

Table 3: Claim-evidence audit. Full details are in `analysis/claim_evidence_audit.csv`.

Claim	Evidence found	Verdict	Required next evidence
Fresh environment and clean replay	Clean replay log installs editable package, runs tests, preflight and convention check; it does not visibly rerun the 25 VMC trajectories.	Partially supported	Add a replay transcript that includes the Stage 4 VMC command or explain why tarball is post-replay output.
Full test suite passed	Submitted log shows 49 pytest dots; local static rerun gives 41 passed, 3 skipped due missing Flax/NetKet.	Supported with caveat	Include explicit pytest summary line; add Stage 4-specific asset tests.
Convention validation passed	Preflight ok; convention JSON ok for N=2/6 checks; Stage 4 rows record machine-precision operator agreement.	Supported	Extend convention JSON to the exact N=8 Stage 4 hard point.
All five matched seeds completed	Raw metrics contain 25 completed rows, five models times five seeds.	Supported	Keep raw schema and retain failed rows in next stages.
Both RBF endpoint candidates passed joint gate	Current `n8_joint_gates.csv` marks endpoint-sum and full endpoint as PASS; stale `n8_joint_gate.csv` conflicts; radial baseline is 0/5 valid.	Supported with caveats	Remove stale file and state radial is a failed control, not a strong baseline.
Endpoint-sum improves all three tracked observables	Median exact paired deltas vs RBF radial are -0.364296, -0.057812, -0.006793, all five seeds lower.	Supported for this hard point	Replicate against adjacent points and capacity-swept controls.
Full endpoint has correlation penalty	Full endpoint energy/density improve, but exact correlation delta is $+0.001956$.	Supported	Treat full endpoint as ablation; prefer endpoint-sum candidate.
No Graph-KAN/N$\geq$10/manuscript yet	Stage 4 output is N=8 chain-only.	Supported	Keep this boundary until controls pass.
Publication-grade result	One point, no stronger controls, no checkpoints, no model generalisation.	Unsupported	Complete Stage 4B/4C before manuscript drafting.

4 Codebase and Reproducibility Audit

4.1 Repository structure

The repository is now a multi-stage production-style research codebase. It includes deterministic geometry, Hamiltonian builders, exact-reference utilities, model code, NetKet runners, observables, statistics, plotting/reporting utilities, stage-specific configs and tests. Stage 4 itself is thin: `scripts/run_stage4.py` delegates to the validated `run_stage3f.py` engine, while `scripts/summarize_stage4.py` adds the corrected dual N=8 gate for endpoint-sum and full endpoint. This is a reasonable reuse choice, but it causes one naming issue: raw Stage 4 files remain named `stage3f_metrics.csv`, `stage3f_site_observables.csv`, and `stage3f_corr_observables.csv`. The README says this is schema compatibility, not relabelling. For public release, use Stage 4 filenames or add an explicit schema-version field.

4.2 Environment, tests, and replay

The submitted Stage 4 preflight records Python 3.12.7 on macOS arm64, CPU-only JAX, JAX 0.7.1, JAXlib 0.7.1, Flax 0.12.0, Optax 0.2.8, NetKet 3.22.3, NumPy 1.26.4 and SciPy 1.13.1. The included conda environment file pins those versions. The SHA-256 of `stage4_n8_five_seed_replication.tar.gz` matches the provided sidecar.

The audit environment lacks Flax, Optax and NetKet, so the expensive NetKet VMC rows were not regenerated. I reran the local static/base tests:

```
cd /mnt/data/stage4_audit_work/extracted/code_pkg/graphkan-amo-nqs-stage4-publication
    -grade-pipeline-fixed-v3
PYTHONPATH=src python -m pytest -vv -ra
# 41 passed, 3 skipped in 3.76s
```

The submitted clean replay log shows a full NQS-environment run with 49 pytest dots at 100 percent, then successful preflight and convention output. It is useful evidence for dependency hygiene, but future logs should include an explicit pytest summary and the actual VMC command.

4.3 Determinism and metadata

The Stage 4 config hash is stored in the run manifest and all raw rows. Each row records profile, model label, model kind, basis family, seed, system size, instance ID, Hamiltonian parameters, samples, SR shift, parameter count, exact and sample observables, operator discrepancy, wall time and sample diversity. This is substantially stronger than an energy-only table.

Remaining reproducibility weaknesses are mostly artifact-level: no checkpoints, stale aggregate, AppleDouble files, Stage 3F filenames in Stage 4 output, and no VMC replay transcript. These should be fixed before any external release.

4.4 Code-quality concern: gate-definition ambiguity

The `classify()` function in `run_stage3f.py` receives sampled density/correlation diagnostics for the health gate, while exact all-state density/correlation errors are also computed and exported. The Stage 4 headline uses exact all-state metrics, and the endpoint-sum result is strong under those exact metrics. Still, the gate definition should not require reading code. Add explicit `sample_health_validity` and `exact_health_validity` columns in the next package.

5 Experimental Design Audit

5.1 Task definition

Stage 4 is a deliberately narrow transfer gate at $N = 8$, $\Omega = 1$, $\Delta = 2$, $C_6 = 4$. It uses one open Rydberg chain, five seeds, 4096 samples, 64 chains, 700 VMC iterations, SR shift 0.1, learning rate 0.0008, gradient clipping, and exact all-state post-training observables for each trained state. This is the right scope for testing whether the Stage 3E endpoint idea survives the first size increase.

5.2 Models and fairness

The five submitted models are:

- `sine_radial_21p`;
- `sine_endpoint_full_21p`;
- `rbf_radial_21p`;
- `rbf_endpoint_sum_21p`;
- `rbf_endpoint_full_21p`.

All report 21 trainable parameters. This is a clean factorial within compact radial/endpoint pair ansatzes. It is not a complete fairness study. A reviewer will ask whether endpoint-sum is better than a modest MLP-pair, endpoint-MLP, smooth/RBF Jastrow, or RBM-like control under matched features and compute. Stage 4 does not answer that.

5.3 Metric choice

The report correctly uses energy, density, and connected-correlation MAE. Energy alone would be insufficient because earlier stages showed that energy can look acceptable while spatial density is wrong. The connected-correlation result is especially important because it distinguished endpoint-sum from full endpoint at $N = 8$.

5.4 Statistical power

Five seeds are enough for a descriptive hard-gate decision, not for broad inferential claims. When all five paired deltas point in the same direction, a two-sided exact sign test gives $p = 0.0625$. The report therefore treats directionality and effect size as strong descriptive evidence, but avoids formal significance language.

6 Quantitative Results Analysis

6.1 Operator validation

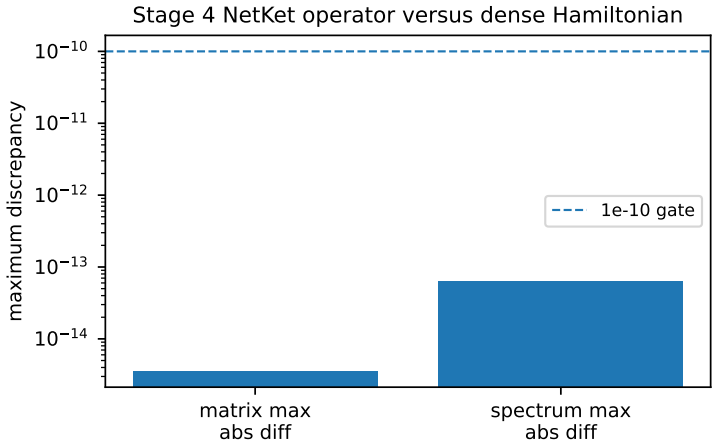


Figure 1: Dense-versus-NetKet operator validation for the submitted Stage 4 rows. The maximum matrix and spectral discrepancies are far below the 10^{-10} gate. This validates the Hamiltonian/operator path for this hard point; it does not validate the ansatz performance claim.

6.2 Health-gate and model summary

Table 4: Stage 4 model summary. Medians are over five seeds. Smaller errors are better. The validity count is the submitted scientific-health label, which is currently based on sampled health diagnostics.

Model	Basis	Params	Valid	$\lvert E - E_0 \rvert$	Density MAE	Corr. MAE	Wall s
RBF endpoint-sum	RBF	21	5/5	0.164078	0.023174	0.020415	19.786
RBF endpoint-full	RBF	21	5/5	0.215474	0.046162	0.029228	19.550
RBF radial	RBF	21	0/5	0.528356	0.081194	0.027209	19.512
sine endpoint-full	sine	21	4/5	0.082828	0.080582	0.038596	19.643
sine radial	sine	21	0/5	0.093236	0.092720	0.043117	19.561

Two points are easy to miss. First, RBF endpoint-sum is not the lowest-energy model in absolute terms; sine endpoint has lower energy but sits on the density-health boundary and has worse correlations. Second, the RBF radial model is an intentionally hard failed control, not a strong baseline. The RBF endpoint-sum result is therefore a repair of a known radial hard-regime failure.

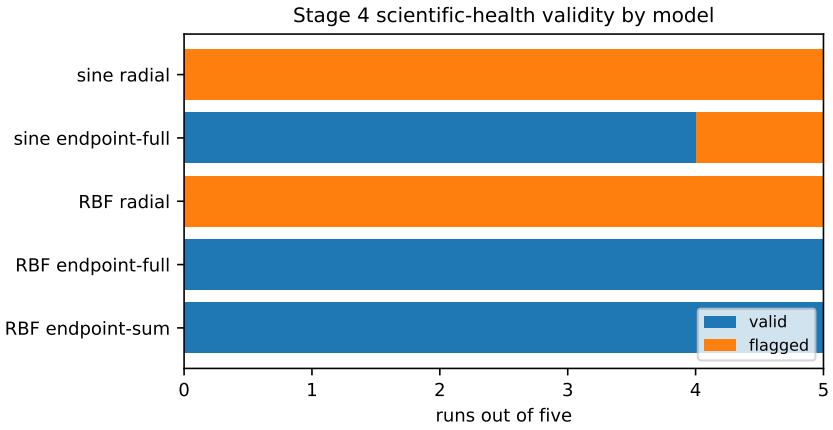


Figure 2: Scientific-health validity by model. RBF endpoint-sum and RBF endpoint-full are 5/5 valid. RBF radial and sine radial are 0/5 valid. Sine endpoint-full is only 4/5 valid and remains close to the density threshold. This supports a targeted endpoint-sum follow-up, not broad scale-up.

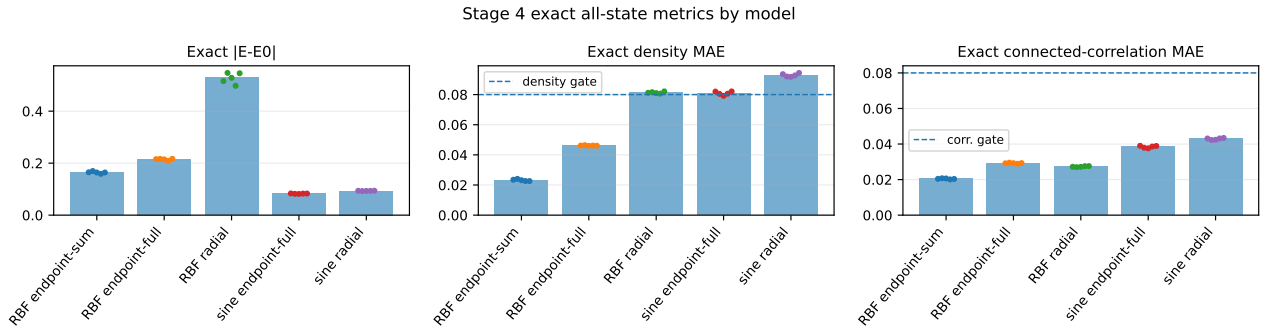


Figure 3: Seed-level exact all-state metrics. Bars show medians and points show seeds. Endpoint-sum is the only submitted RBF variant that improves energy, density, and correlation relative to RBF radial. The sine endpoint model improves over sine radial but remains density-limited.

6.3 Paired endpoint effects

Table 5: Key paired candidate-minus-baseline deltas. Negative is better for all three error metrics. Bootstrap intervals are descriptive because $n = 5$ paired seeds.

Comparison	Metric	Pairs	Median delta	Bootstrap low	Bootstrap high	Lower fraction	Sign p
RBF endpoint-sum minus RBF radial	Energy	5	-0.364296	-0.381758	-0.338682	1.0	0.0625
RBF endpoint-sum minus RBF radial	Density	5	-0.057812	-0.059506	-0.057503	1.0	0.0625
RBF endpoint-sum minus RBF radial	Corr.	5	-0.006793	-0.007444	-0.006246	1.0	0.0625
RBF endpoint-full minus RBF radial	Energy	5	-0.313732	-0.330781	-0.288721	1.0	0.0625
RBF endpoint-full minus RBF radial	Density	5	-0.035025	-0.036046	-0.034464	1.0	0.0625
RBF endpoint-full minus RBF radial	Corr.	5	+0.001956	+0.001361	+0.002530	0.0	0.0625
RBF endpoint-sum minus endpoint-full	Energy	5	-0.050563	-0.052590	-0.046617	1.0	0.0625
RBF endpoint-sum minus endpoint-full	Density	5	-0.022829	-0.023518	-0.022356	1.0	0.0625
RBF endpoint-sum minus endpoint-full	Corr.	5	-0.008776	-0.008924	-0.008591	1.0	0.0625

The paired table is the core Stage 4 result. Endpoint-sum is not merely a relabelling of the full endpoint variant: it is better than full endpoint on all three exact metrics in all five seeds. This makes endpoint-sum the correct next candidate.

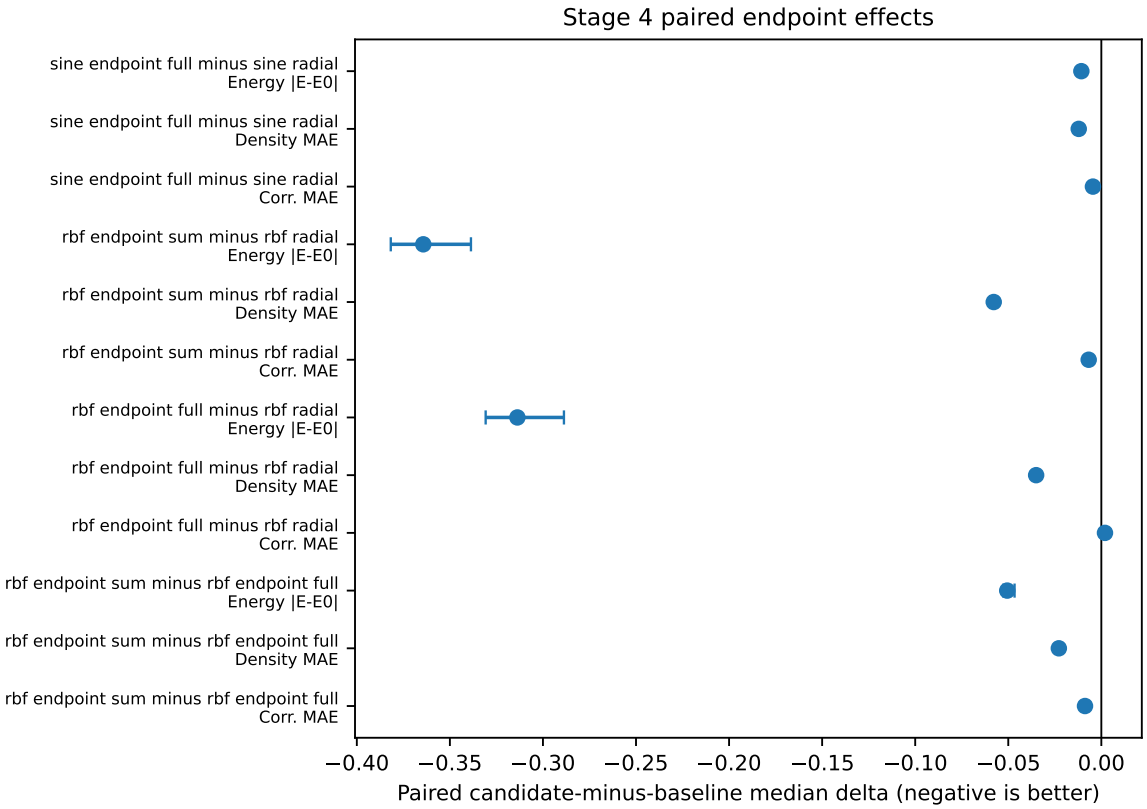


Figure 4: Paired endpoint effects. Negative values favour the endpoint candidate. The endpoint-sum versus RBF radial deltas are strongly negative on energy, density and connected correlation. Full endpoint improves energy and density but has a correlation penalty.

6.4 Relative improvements

Against RBF radial, RBF endpoint-sum reduces the median exact energy error by approximately 69 percent, density MAE by 71 percent, and connected-correlation MAE by 25 percent. Full endpoint reduces energy and density errors by approximately 59 and 43 percent respectively, but increases the median connected-correlation MAE by about 7 percent. These are practical effect sizes at this single point, not general method-level effect sizes.

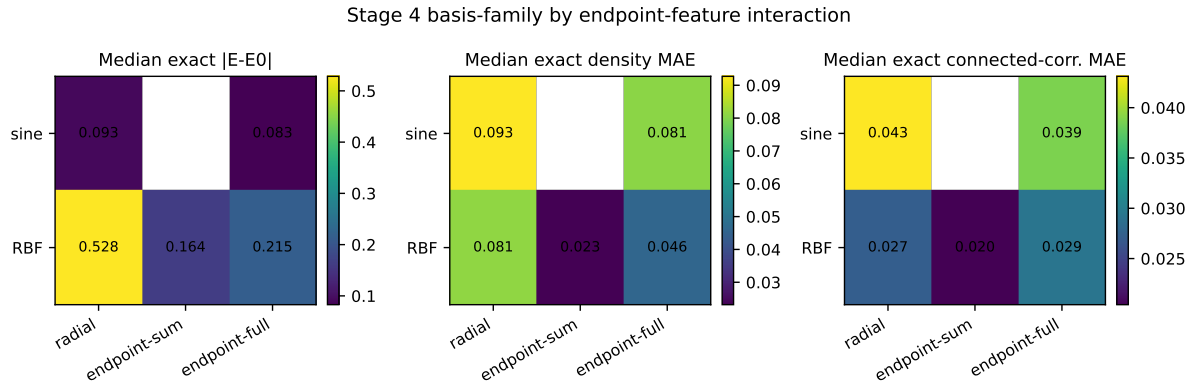


Figure 5: Basis-family by endpoint-feature interaction. The RBF endpoint-sum cell is the only cell that combines low density error and improved correlation. The sine endpoint cell improves over sine radial but remains near the density threshold. This supports a basis-feature interaction interpretation, not generic KAN superiority.

7 Negative-Result Autopsy and Failure Diagnosis

Stage 4 contains positive evidence, but the most useful scientific information is still in the weak and failed cases. These failures constrain the interpretation and define the next experiments.

Table 6: Negative-result autopsy and rescue hypotheses.

Weak or negative result	Possible cause	Evidence supporting this cause	Diagnostic test	Cost	Priority
RBF radial is 0/5 valid	Pure radial features cannot encode open-chain boundary environment at fixed capacity	Endpoint-sum reduces density MAE from 0.0812 to 0.0232 and repairs site-region errors	Capacity-sweep radial RBF versus endpoint-sum with longer training	Tier 1–2	High
Sine radial is 0/5 valid and sine endpoint is 4/5 valid	Sine/Fourier basis may not fit hard-regime density structure	Sine endpoint improves all metrics but remains near the density gate	Basis sweep: sine, RBF, Chebyshev, spline with endpoint-sum	Tier 1	Medium-high
Full endpoint worsens correlation relative to RBF radial	Extra endpoint difference/product features may distort pair correlations	Endpoint-sum beats full endpoint on all exact metrics	Component ablation sum/diff/product/full at N=6 and N=8	Tier 1	High
MLP controls absent from Stage 4 table	Stage 4 tested transfer before capacity controls	Config contains no MLP rows, only compact KAN/Jastrow variants	Width/depth and compute-swept MLP radial/endpoint controls	Tier 2	Critical
Health gate mixes sample and exact metrics	Classification uses sample density/correlation, while report emphasizes exact observables	Raw rows expose both sample and exact metrics; code confirms classify uses sample metrics	Add dual validity columns and exact gate output	Tier 0	High
Stale aggregate file conflicts	Legacy single-gate file retained after v3 summary correction	`n8_joint_gate.csv` is NOT_APPLICABLE while `n8_joint_gates.csv` passes	Remove/deprecate stale file; add schema test	Tier 0	Medium

True negative or false negative? The radial and sine failures should not yet be called true negatives. They may reflect real ansatz-form limitations, but they could also reflect basis scaling, insufficient capacity, optimiser budget, or gate-definition choices. The full-endpoint correlation penalty is more informative: adding all endpoint features is not automatically better, and the minimal endpoint-sum feature is empirically better in the submitted N=8 gate.

8 Repair Hypotheses and Salvage Plan

Table 7: Repair hypothesis matrix.

Hypothesis	Code change	Experiment	Failure criterion	Resource cost	Decision impact
Endpoint-sum is the minimal boundary-aware feature	Keep endpoint-sum as first-class model; export endpoint features and coefficients	N=6 and N=8 component ablation: sum/diff/product/full/radial	Difference/product/full dominate or no endpoint variant transfers	Tier 1	Focus paper on simple reflection-symmetric conditioning
Result is not tiny-model artifact	Add basis/training capacity sweep and compute manifest	N=8 radial and endpoint-sum with $n_{basis} \in \{8, 12, 16, 24\}$ and longer training	Radial catches up after modest capacity increase	Tier 2	Separates representation effect from underpowered radial control
Matched MLP may reproduce endpoint-sum	Implement MLP width/depth/activation and endpoint features	MLP radial/endpoint at 21p, 64p, 128p and compute-matched budgets	Modest MLP endpoint dominates endpoint-sum	Tier 2	Determines whether claim is boundary-feature-specific rather than KAN-specific
N=8 result generalises locally	Add adjacent hard-regime configs	$\Delta \in \{1.5, 2.0, 2.5\}$, $C_6 \in \{3, 4, 5\}$, $N \in \{6, 8\}$	Effect appears only at one point	Tier 2–3	Enables a publishable regime-map story
Feature helps size transfer	Add N=10/12 chain configs with checkpointing	Train/evaluate endpoint-sum and controls on N=10/12	Benefit disappears at first larger size	Tier 3	Gate to finite-size story

What would make the current result publishable?

The current result becomes a publishable positive result only if endpoint-sum survives three filters: (1) it remains non-dominated by MLP/Jastrow controls under matched features and compute; (2) it works across an adjacent hard-regime parameter patch rather than only at one point; and (3) its site/correlation diagnostics support a clear physical mechanism rather than boundary memorisation. Without these filters, the result is still useful but should remain an internal diagnostic or become part of a negative/repair-map paper.

9 Robustness, Ablation, and Diagnostic Analysis

9.1 Site-resolved density diagnostic

The most informative Stage 4 diagnostic is site-resolved density. The ED density profile at the hard point is edge-high, near-edge-low, inner-peak-high, central-intermediate. RBF radial misses near-edge and central structure. RBF endpoint-sum reduces edge and near-edge errors while keeping central errors below the global density threshold.

Table 8: Median exact density absolute error by chain region. Regions are edge sites 0/7, near-edge sites 1/6, inner-peak sites 2/5, and central sites 3/4.

Model	Edge	Near-edge	Inner-peak	Central
RBF endpoint-sum	0.011037	0.007037	0.037593	0.036974
RBF endpoint-full	0.046913	0.032416	0.075052	0.030403
RBF radial	0.064871	0.143180	0.023270	0.093285
sine endpoint-full	0.116965	0.102750	0.054843	0.049831
sine radial	0.136733	0.114093	0.064730	0.056748

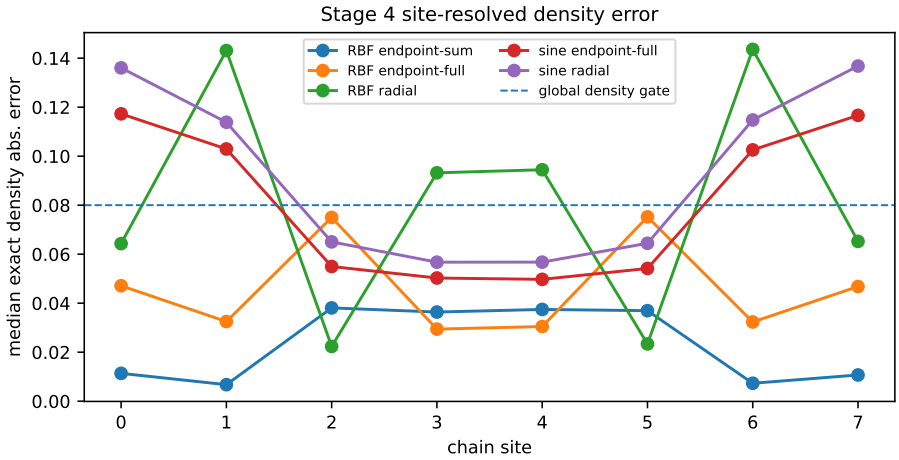


Figure 6: Site-resolved density error. RBF endpoint-sum dramatically reduces near-edge errors relative to RBF radial. Full endpoint improves some regions but remains worse than endpoint-sum, especially at the inner-peak sites. This motivates endpoint-component ablations.

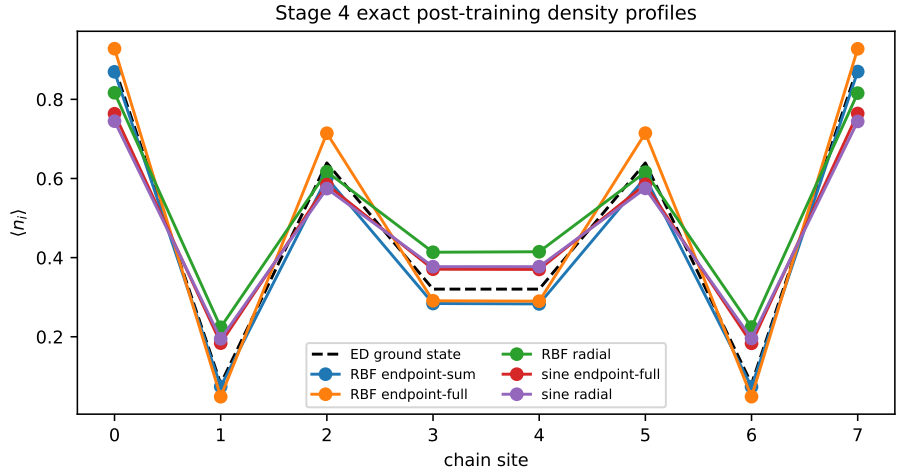


Figure 7: Exact post-training model density profiles compared with ED. RBF endpoint-sum is closest among submitted models to the qualitative ED profile. This supports the boundary-aware repair interpretation at this one hard point. It does not prove transfer to other geometries.

9.2 Connected-correlation diagnostic

The correlation evidence is important because it changes the preferred model. At Stage 3E, full endpoint improved density but worsened correlation relative to RBF radial. At Stage 4, endpoint-sum improves correlation relative to both RBF radial and full endpoint. The median per-entry correlation-error summary is:

Table 9: Connected-correlation absolute-error summary from exact all-state observables.

Model	Median entry error	Mean entry error	Max median entry error
RBF endpoint-sum	0.013269	0.019081	0.050322
RBF radial	0.025599	0.029378	0.099740
RBF endpoint-full	0.026138	0.028832	0.067953
sine endpoint-full	0.037896	0.039176	0.075690
sine radial	0.042566	0.043733	0.085499

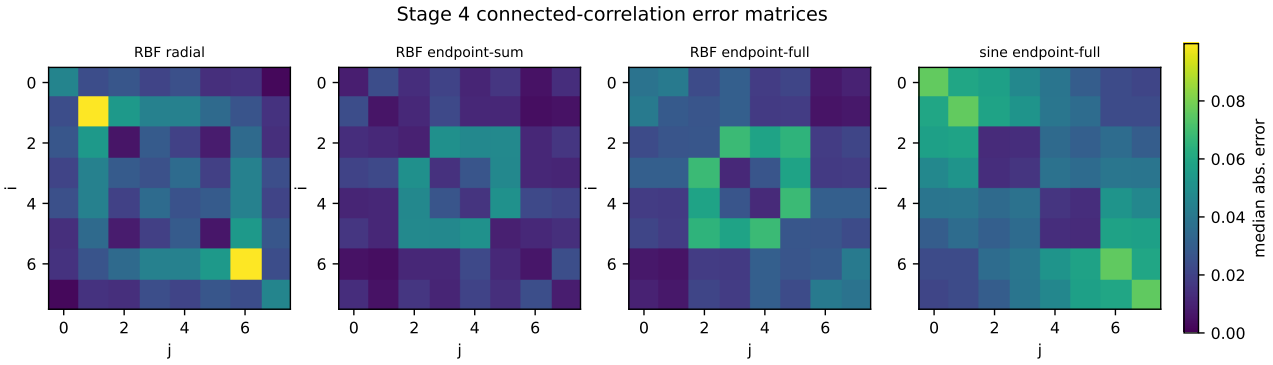


Figure 8: Connected-correlation error matrices. Endpoint-sum lowers the scalar correlation MAE and the largest median matrix-entry errors compared with RBF radial and full endpoint. This is the main reason endpoint-sum, not full endpoint, should be the next candidate.

9.3 Learning curves

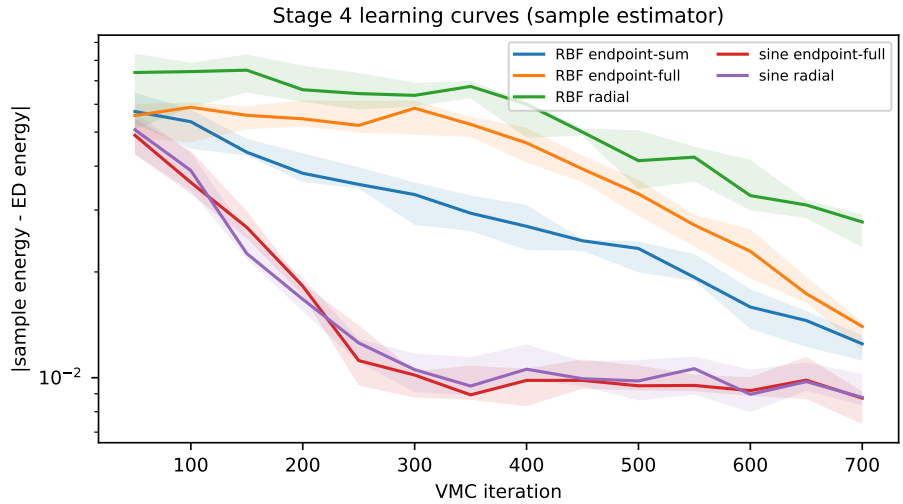


Figure 9: Sample-energy learning curves. These curves show that all models train in the submitted VMC loop, but energy-only curves would not distinguish the density/correlation failures. Exact post-training observables remain the deciding metrics.

10 Figures and Visual Interpretation

The Stage 4 figures support one local story: endpoint-sum repairs the N=8 RBF radial hard-point failure more cleanly than full endpoint. They do not support manuscript claims about generic KAN performance, graph models, transfer, or top-journal readiness. Figures 10 and 11 place the result in context.

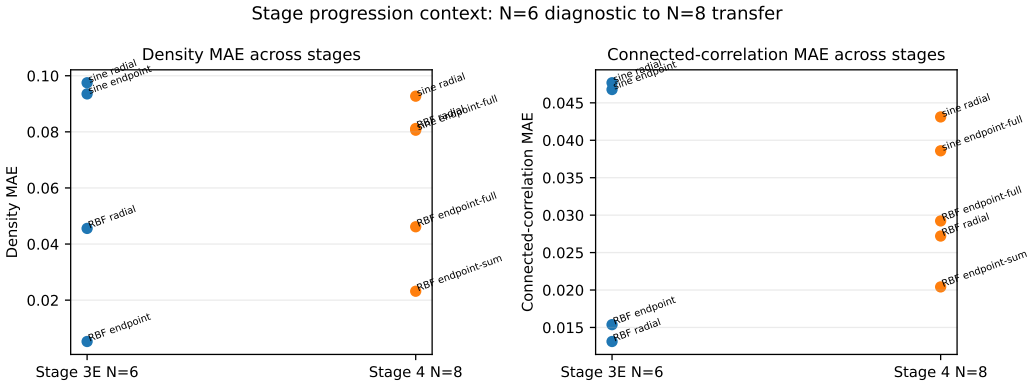


Figure 10: Stage progression context from the prior N=6 diagnostic to the N=8 transfer gate. The trend is encouraging because the N=8 endpoint-sum variant improves the correlation behavior that was ambiguous for full endpoint at N=6. The figure is internal-only until controls are added.

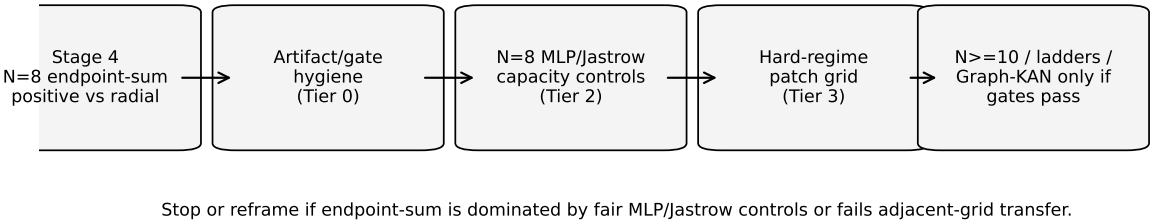


Figure 11: Recommended decision flow. Stage 4 authorises artifact cleanup and control sweeps, not direct Graph-KAN or N≥10 scale-up. The bold route is conditional on endpoint-sum surviving stronger controls and adjacent-grid transfer.

11 Ambitious Experiment Ladder Toward Publishability

A conservative current verdict should not lead to a timid research plan. The correct strategy is a gated escalation ladder: cheap repairs first, then bolder campaigns once the artifact and fairness gates are clean.

Table 10: Ambitious experiment ladder.

Tier	Experiment	Objective	Hardware	Runtime	Claim unlocked if it passes
0	Artifact hygiene and gate-regression checks	Remove stale files; add exact/sample health columns; validate schemas and hashes	CPU laptop	Minutes	Stage 4 artifact is internally auditable
1	N=8 endpoint-component replay	Confirm sum vs diff/product/full/radial with exact observables	CPU/MPS/GPU	1–4 h	Endpoint-sum selected as hard-regime repair candidate
2	N=8 MLP and smooth/RBF control sweep	Test whether boundary feature, not KAN branding, drives result	Laptop GPU or overnight CPU	Overnight	Fair small-model boundary-feature result

Tier	Experiment	Objective	Hardware	Runtime	Claim unlocked if it passes
3	Hard-regime patch validation	Check N=6/8 and adjacent Δ, C_6 neighborhood	Team machines/Colab chunks	1–3 days	Rydberg hard-regime repair map
4	Publication-grade chain/lattice campaign	Size transfer, ladders, open/periodic controls, strong baselines	RTX 2060–3080 Ti class or CPU chunks	1–3 weeks	Specialist-journal candidate
5	Optional external/tensor campaign	Add DMRG/tensor references or larger baselines if Tier 4 warrants it	Optional external/HPC	Conditional	Stretch-venue plausibility, not guaranteed

Bold experiments include: the Stage 4B capacity/compute MLP control sweep; the contiguous hard-regime patch grid; the N=10/12 finite-size chain campaign; the open-versus-periodic/ladders boundary-generalisation campaign; and, only after these pass, Graph-KAN geometry transfer against a graph MPNN baseline.

12 Resource-Aware Compute Plan

Table 11: Resource-aware plan under laptop/team-machine constraints.

Experiment	Tier	CPU/GPU	VRAM	Runtime	Checkpoint and waste control
Artifact cleanup and dual-gate tests	0	CPU only	None	15–30 min	No expensive compute until schema tests pass
N=8 component replay	1	CPU ok; GPU useful	$\leq$6 GB	1–4 h	Per-run CSV checkpoints; run seed/model parallel
N=8 MLP/Jastrow control sweep	2	GPU recommended; CPU overnight possible	6–12 GB	8–24 h	Chunk by model/seed; stop if baseline clearly dominates
N=6/8 hard-regime grid	3	Team machines or Colab chunks	6–12 GB	1–3 days	Hamiltonian x seed x model parallel; resume by missing run IDs
N=10/12 chain transfer	3–4	GPU VMC; CPU ED/DMRG where feasible	8–16 GB	Several days to weeks	Launch only after control gate; checkpoint every chunk
Graph-KAN geometry transfer	4	GPU recommended	12–16 GB	1–3 weeks	Conditional on pair-model success; compare to MPNN

Use CPU for Tier 0 and small exact checks. Use laptop GPU/MPS/Colab Free for Tier 1–3 VMC chunks. Do not launch Tier 4 unless Stage 4B and Stage 4C pass; otherwise compute is likely wasted.

13 Path to a Top-Journal-Quality Result

13.1 Claim architecture

Table 12: Publication claim ladder.

Current honest claim	After repair	After medium validation	Forbidden claim
At one N=8 hard open-chain point, RBF endpoint-sum improves exact energy, density, and correlation errors against matched RBF radial in five paired seeds.	Endpoint-sum is the selected boundary-aware RBF pair model after component and gate-hygiene repairs.	Endpoint-sum repairs a contiguous hard-regime patch and survives MLP/Jastrow controls.	KANs or Graph-KANs are superior based on Stage 4 alone.
Full endpoint is not preferred because it retains a correlation penalty.	Full endpoint becomes an ablation showing more endpoint features are not always better.	Endpoint-feature minimality can become a mechanism insight.	Endpoint features universally improve all observables.

13.2 Current strongest possible honest claim

The strongest claim today is: *At the registered $N = 8$ hard open-chain point, a 21-parameter RBF endpoint-sum pair model repairs the RBF radial hard-regime failure on exact energy, density, and connected-correlation metrics across five paired seeds.*

13.3 Strongest claim after minimal repair

After artifact cleanup and endpoint-component replay, the claim could become: *The relevant hard-regime correction is the reflection-symmetric endpoint-sum feature rather than unrestricted endpoint features.*

13.4 Strongest claim after medium-scale validation

After MLP/Jastrow controls and adjacent hard-regime validation, the claim could become: *Boundary-aware pair features provide a compact, interpretable correction for strong-interaction open-chain Rydberg hard regimes, remaining non-dominated by matched MLP/Jastrow controls.*

13.5 What would make the work top-journal worthy

The work becomes top-journal plausible only if the endpoint/geometry-aware mechanism generalises beyond one chain point: across sizes, adjacent parameters, boundary conditions, ladders/irregular arrays, and against strong graph/MLP/autoregressive controls. A new physical diagnostic - for example, a gauge-fixed learned pair function that reliably tracks blockade/correlation structure - would be more important than a small energy improvement.

13.6 Fallback routes

If endpoint-sum survives controls but only on chains, target a solid specialist paper in Physical Review A or MLST. If it fails against fair MLP controls, pivot to a negative-results benchmark/failure-map paper: useful, but not a method-success paper. If the software remains hard to reproduce or gate definitions stay ambiguous, do not publish externally.

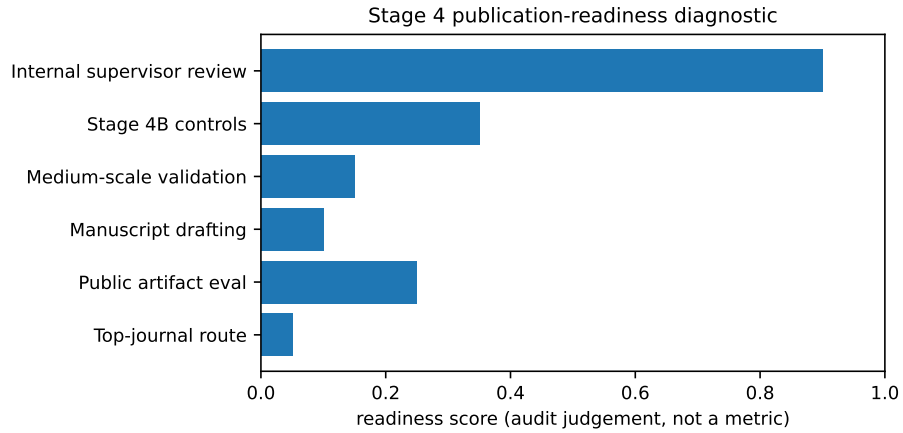


Figure 12: Publication-readiness diagnostic. Scores are supervisory judgements, not measured performance. Stage 4 is ready for internal review, not manuscript drafting or public artifact evaluation.

13.7 Reviewer objection map

Table 13: Likely reviewer objections and required responses.

Objection	Why reviewer objects	Current weakness	Required evidence
RBF radial is not a strong baseline	It fails the health gate	Stage 4 compares endpoint variants mainly to a failed control	Capacity-swept MLP/Jastrow/RBM or reduced RBM controls
One hand-picked point	The point was selected after failure-map discovery	No adjacent parameter or larger-N Stage 4 result	Frozen adjacent hard-regime grid and N=10/12 transfer
Endpoint features memorize boundaries	Open-chain endpoint coordinate encodes site location	No periodic/ladder/irregular tests	Open/periodic control, ladder/irregular transfer, region-wise errors
Gate logic ambiguous	Sample validity and exact metrics differ	`classify()` uses sampled density/correlation	Separate exact and sample health-gate columns
No checkpoints	Artifact evaluator cannot recompute observables	Only CSV outputs included	Saved Flax/NetKet parameters or deterministic replay artifacts

14 Risks, Weaknesses, and Failure Modes

Main supervisor risk

The danger is to treat a strong repair against a failed radial control as a method-level result. The opposite danger is to dismiss it as too small. The correct middle path is to convert it into a clean mechanism test: endpoint-sum may be a compact boundary-aware correction for the open-chain hard regime, but that mechanism must survive fair controls and adjacent-grid transfer.

- **Methodological risk:** endpoint-sum may be beaten by a modest endpoint-MLP or smooth/RBF Jastrow after a fair capacity sweep.

- **Statistical risk:** five seeds are useful but not enough for formal claims; one Hamiltonian point is not generalisation.
- **Software risk:** stale aggregate files and dual sample/exact gate definitions could confuse future auditors.
- **Reproducibility risk:** no model checkpoints and no visible full VMC replay in the clean replay log.
- **Publication risk:** reviewers will reject KAN-superiority language; the story must be boundary-feature/benchmark oriented.
- **Resource risk:** Graph-KAN or N≥10 too early could waste compute before cheaper controls answer the main uncertainty.
- **Strategic risk:** being too conservative could miss a real mechanism. If controls pass, the project should escalate boldly.

15 Next-Iteration Plan

15.1 Stage name

Stage 4B: N=8 Hard-Point Fairness, Gate-Definition, and Endpoint-Sum Control Sweep.

15.2 Objective

Determine whether endpoint-sum remains non-dominated once the artifact is clean, the gate definition is explicit, and matched MLP/Jastrow controls are run under the same N=8 hard point.

15.3 Required code changes

1. Replace or deprecate `aggregated/n8_joint_gate.csv`; keep only the current dual-gate file.
2. Add `sample_health_validity` and `exact_health_validity` fields.
3. Add `tests/test_stage4_assets.py` checking config, raw schemas, dual gates, and no stale contradictory gate files.
4. Save trained parameters/checkpoints for at least endpoint-sum, RBF radial, and the strongest MLP/Jastrow controls.
5. Rename Stage 4 raw outputs or add a clear `schema_version` field explaining Stage 3F compatibility.

15.4 Required experiments

1. **Endpoint component replay at N=8:** radial, endpoint-sum, endpoint-diff, endpoint-product, endpoint-full, five seeds, exact observables.
2. **MLP control sweep:** MLP-pair and endpoint-MLP at 21, 64, and 128 parameter scales, plus compute-matched budgets.
3. **Smooth/RBF Jastrow controls:** distance-binned, smooth/RBF, and endpoint-sum analogues if implementable.
4. **Gate comparison output:** exact and sample gates, paired bootstrap tables, site-region errors, correlation matrices, runtime/parameter tables.

15.5 Command template

```
conda env create -f environment-nqs-py312-macos-arm64-stage4.yml
conda activate graphkan-stage4-py312
PYTHONPATH=src python -m pytest -q -ra
PYTHONPATH=src python scripts/preflight_nqs.py
PYTHONPATH=src python scripts/check_netket_conventions.py \
  --config configs/netket_stage2_6_conventions.yaml
PYTHONPATH=src python scripts/run_stage4b_controls.py \
  --config configs/netket_stage4b_n8_controls.yaml --overwrite
PYTHONPATH=src python scripts/summarize_stage4b.py \
  --results results_stage4b/n8_controls
```

15.6 Pass/fail logic

Pass Stage 4B if endpoint-sum is not dominated on energy, density, correlation and compute by the best matched MLP/Jastrow control, and if the dual exact/sample gate definitions agree for the headline rows. **Proceed to Stage 4C** only after Stage 4B passes. **Stop scale-up** if endpoint-sum is dominated by a modest endpoint-MLP or smooth/RBF Jastrow under matched budget. **Reframe** if endpoint-sum helps density but hurts correlations in adjacent settings: the story becomes a diagnostic failure-map paper, not a method paper.

16 Supervisor Checklist

Table 14: Checklist to send back to the student.

Category	Items
Must fix	Remove stale `n8_joint_gate.csv`; separate sample and exact health gates; add Stage 4 asset/schema tests; run N=8 MLP/Jastrow capacity controls.
Should fix	Save model checkpoints; include explicit pytest summary; rename Stage 4 raw files or add schema version; extend convention check to N=8 hard point.
Optional improvements	Chebyshev/spline basis sweep; endpoint-feature coefficient export; site-region automated diagnostics.
Do not do yet	No Graph-KAN, no geometry transfer, no $N \geq 10$, no manuscript drafting, no top-journal language.
Allowed claims	Endpoint-sum improves all three exact observables over matched RBF radial at one N=8 hard point in five paired seeds.
Forbidden claims	KAN superiority, Graph-KAN value, geometry transfer, general Rydberg benchmark success, publication readiness.
Conditional claims	Boundary-aware pair-feature mechanism only after MLP/Jastrow controls and adjacent hard-regime grid pass.
Files to send next	Clean code ZIP, result tarball with hash, raw metrics, exact site/correlation observables, paired bootstrap table, model checkpoints, clean replay log including VMC command, figure scripts.

17 Final Recommendation

Current evidence decision. Stage 4 is a real positive diagnostic: endpoint-sum repairs the RBF radial hard-point failure at $N = 8$ across five seeds and improves all three exact observables relative to that radial control. Accept it as a meaningful internal result.

Next safe action. Run Stage 4B controls and artifact cleanup. The most important next experiment is the N=8 capacity/compute control sweep for endpoint-MLP, MLP-pair, smooth/RBF Jastrow, RBF radial, full endpoint, and endpoint-sum.

Bold route if gates pass. If Stage 4B passes, launch the hard-regime patch grid across $N \in \{6, 8\}$, adjacent Δ and C_6, then N=10/12 finite-size transfer and ladder/open-boundary tests. This uses available laptop/GPU/Colab resources sensibly without jumping directly into Graph-KAN.

Fallback route if gates fail. If a fair MLP/Jastrow control dominates endpoint-sum, do not hide the result. Reframe the project as a rigorous Rydberg NQS failure-map/benchmark study showing when compact radial and boundary-aware pair ansatzes fail or succeed.

Publication strategy. The present data are not a manuscript. The plausible paper after successful gates is a specialist-journal Rydberg simulation/MLST benchmark paper centered on compact boundary-aware pair functions, exact observables, fairness controls, and failure maps. A top-journal path requires a generalisable physical diagnostic or transfer principle; Stage 4 alone does not provide that.

A Full File Inventory

The full inventory is supplied as `analysis/full_file_inventory.csv`. The summary is in `analysis/file_inventory_summar`
The result archive contains 21 AppleDouble/platform files, which should be removed before public release.

B Additional Analysis Files

The report project includes the following generated analysis files:

- analysis/stage4_model_summary_detailed.csv
- analysis/stage4_paired_deltas_bootstrap.csv
- analysis/stage4_relative_changes_vs_rbf_radial.csv
- analysis/stage4_sample_vs_exact_health_metrics.csv
- analysis/stage4_site_region_density_errors.csv
- analysis/stage4_corr_error_summary.csv
- analysis/negative_result_autopsy.csv
- analysis/repair_hypothesis_matrix.csv
- analysis/ambitious_experiment_ladder.csv
- analysis/resource_aware_compute_plan.csv
- analysis/reviewer_objection_map.csv

C Reproducibility Commands

```
# Audit extraction and verification
unzip -q /mnt/data/graphkan-amo-nqs-stage4-publication-grade-pipeline-fixed-v3.zip -d /mnt/data/
    stage4_audit_work/extracted/code_pkg
tar -xzf /mnt/data/stage4_n8_five_seed_replication.tar.gz -C /mnt/data/stage4_audit_work/extracted/
    results_pkg
sha256sum /mnt/data/stage4_n8_five_seed_replication.tar.gz

# Static tests in review environment
cd /mnt/data/stage4_audit_work/extracted/code_pkg/graphkan-amo-nqs-stage4-publication-grade-pipeline-fixed-
    v3
PYTHONPATH=src python -m pytest -vv -ra

# Aggregation verification
PYTHONPATH=src python scripts/summarize_stage4.py \
  --results /mnt/data/stage4_audit_work/extracted/results_pkg/results_stage4/n8_five_seed_replication
```

D Environment Notes

Submitted preflight: Python 3.12.7, macOS arm64, JAX 0.7.1, JAXlib 0.7.1, Flax 0.12.0, Optax 0.2.8, NetKet 3.22.3, NumPy 1.26.4, SciPy 1.13.1, CPU-only JAX device. Review environment: Python 3.13.5, no Flax/Optax/NetKet; NetKet VMC was not rerun locally.

E Suggested Message to the Student

> Subhojit, Stage 4 is a useful and encouraging hard-point transfer result. I approve it as an internal diagnostic: RBF endpoint-sum improves exact energy, density, and connected-correlation error against matched RBF radial at $N = 8$, $\Delta = 2$, $C_6 = 4$ across five seeds. Do not describe this as KAN superiority or as publication-grade evidence. The next step is Stage 4B, not N≥10 or Graph-KAN: clean the stale aggregate files, separate sample and exact health gates, add Stage 4 schema tests, save checkpoints, and run the N=8 MLP/Jastrow capacity and compute controls. If endpoint-sum survives those controls, we will approve a hard-regime patch grid and finite-size transfer ladder. If it does not, we will reframe the project as a rigorous benchmark/failure-map study rather than a method-success paper.

F Next Student Response Template

The next response should contain:

1. Short update note: what changed and which gate was targeted.

2. Code diff summary: files changed and why.
3. Exact commands run, including environment creation, tests, preflight, convention checks, VMC, summarisation.
4. Test results with explicit pytest summary.
5. Output archive plus SHA-256 sidecar.
6. Raw metrics, exact site observables, exact correlation observables, model checkpoints, aggregate tables.
7. Generated figures and the scripts that produced them.
8. Known failures and all flagged rows.
9. Interpretation separating pipeline success, model performance, fairness controls, and generalisation.
10. Questions for supervisor, limited to decisions that the data actually support.

References

[1] Prior uploaded report, *Scientific Audit and Next-Iteration Review of GraphKAN AMO NQS: Stage 3E Matched-Fairness Replication*, 26 June 2026.

[2] Uploaded Stage 4 student prompt and supervision request, `Pasted text(152).txt`, 26 June 2026.